

The Derivative

Complex Analysis

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Autumn 2026

Definition

Let Ω be an open subset of $\mathbb{C}$, let $f : \Omega \rightarrow \mathbb{C}$ and let $z_0 \in \Omega$. We say that a complex number L is the **derivative of f at z_0** if given any $\varepsilon > 0$ there exists $\delta > 0$ such that if $z \in \Omega$ satisfies $0 < |z - z_0| < \delta$, then

$$\left| \frac{f(z) - f(z_0)}{z - z_0} - L \right| < \varepsilon.$$

In this case we say that f is **complex differentiable** (or simply **differentiable**) at z_0 , and write $f'(z_0)$ for L . If f is differentiable at each point of its domain Ω , then we say that f is **differentiable**.

Definition

Let Ω be an open subset of $\mathbb{C}$, let $f : \Omega \rightarrow \mathbb{C}$ and let $z_0 \in \Omega$.

If f is differentiable at each point of some neighborhood of z_0 , then f is said to be **holomorphic** at z_0 and if f is holomorphic at each point of an open subset Φ of Ω , then f is said to be **holomorphic** on Φ .

We say that f is **entire** if it is holomorphic on $\mathbb{C}$. Furthermore, the family of all holomorphic functions on Ω will be denoted by $\mathcal{H}(\Omega)$.

Definition

A point z_0 is called a **singularity** (or a **singular point**) of f if f is not holomorphic at z_0 but for any given $\varepsilon > 0$, f is holomorphic at some point of $B_\varepsilon(z_0)$

Definition

Let Ω be an open subset of $\mathbb{C}$ and let $f : \Omega \rightarrow \mathbb{C}$. If f' exists and has a derivative at z_0 , then we say that f is **twice differentiable at z_0** , and we refer to the resulting number as the **second derivative of f at z_0** which we will denote by $f''(z_0)$ or by $f^{(2)}(z_0)$. In a similar fashion we define the **third derivative $f^{(3)}(z_0)$, ..., and the n th derivative $f^{(n)}(z_0)$ of f at z_0** , whenever these derivatives exist. If $f^{(n)}(z_0)$ exists for all $n \in \mathbb{N}$, then we say that f is **infinitely differentiable at z_0** and if f is infinitely differentiable on Ω , then we simply say that f is **smooth (or infinitely differentiable)**.

Definition

Let Ω be a subset of $\mathbb{C}$, let $f : \Omega \rightarrow \mathbb{C}$, let $k \in \mathbb{N}$ and let Λ be an open subset of Ω . We say that f is **class C^k on Λ** if the derivatives up to order k all exist and are continuous (including the k th derivative) on Λ , and write $f \in C^k[\Lambda]$. If f is class C^k on Ω , then we simply say that f is **class C^k** , and write $f \in C^k$. We define $C[\Lambda] \equiv C^0[\Lambda]$ to be the set of all continuous functions on Λ . Further, if f is infinitely differentiable at each point of Λ , then we say that f is **class C^∞ on Λ** and write $f \in C^\infty[\Lambda]$.

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Theorem 1.1

If $f : \Omega \rightarrow \mathbb{C}$ is differentiable at $z_0 \in \Omega$, then f is continuous at z_0 .

Theorem 1.2

Let Ω be an open subset of $\mathbb{C}$, let $z_0 \in \Omega$ and let $f : \Omega \rightarrow \mathbb{C}$ and $g : \Omega \rightarrow \mathbb{C}$ be functions that are differentiable at z_0 . Then:

i) If $\lambda \in \mathbb{C}$, then the function λf is differentiable at z_0 , and

$$(\lambda f)'(z_0) = \lambda f'(z_0).$$

ii) The function $f + g$ is differentiable at z_0 , and

$$(f + g)'(z_0) = f'(z_0) + g'(z_0).$$

iii) *The function fg is differentiable at z_0 , and*

$$(fg)'(z_0) = f'(z_0)g(z_0) + f(z_0)g'(z_0).$$

iv) *If $g(z_0) \neq 0$, then the function f/g is differentiable at z_0 , and*

$$\left(\frac{f}{g}\right)'(z_0) = \frac{f'(z_0)g(z_0) - f(z_0)g'(z_0)}{(g(z_0))^2}.$$

Corollary 1.3

Let Ω be an open subset of $\mathbb{C}$, let $z_0 \in \Omega$, let $f : \Omega \rightarrow \mathbb{C}$ and $g : \Omega \rightarrow \mathbb{C}$ be functions such that f is in $\mathcal{H}(\Omega)$ and g is in $\mathcal{H}(\Omega)$. Then:

- i) If $\lambda \in \mathbb{C}$, then the function λf is in $\mathcal{H}(\Omega)$.
- ii) The function $f + g$ is in $\mathcal{H}(\Omega)$.
- iii) The function fg is in $\mathcal{H}(\Omega)$.
- iv) If $g(z) \neq 0$ for all $z \in \Omega$, then the function f/g is in $\mathcal{H}(\Omega)$.

Theorem 1.4 (Carathéodory's Theorem)

Let f be defined on an open subset Ω of $\mathbb{C}$ containing the point z_0 . Then f is differentiable at z_0 if and only if there exists a function φ on Ω that is continuous at z_0 and satisfies

$$f(z) - f(z_0) = \varphi(z)(z - z_0) \quad \text{for } z \in \Omega.$$

In this case, we have $\varphi(z_0) = f'(z_0)$.

Example 1.5

Define $f(z) = z^3$ for $z \in \mathbb{C}$. Show that f is differentiable by showing the existence of a function φ as in Carathéodory's theorem.

Theorem 1.6 (The Chain Rule)

Let Ω, Ψ be open subsets of $\mathbb{C}$, let $g : \Omega \rightarrow \mathbb{C}$ and $f : \Psi \rightarrow \mathbb{C}$ be functions such that $f(\Psi) \subseteq \Omega$, and let $z_0 \in \Psi$. If f is differentiable at z_0 and if g is differentiable at $f(z_0)$, then the composite function $g \circ f$ is differentiable at z_0 and

$$(g \circ f)'(z_0) = (g' \circ f)(z_0) \cdot f'(z_0).$$

Corollary 1.7

Let Ω, Ψ be open subsets of $\mathbb{C}$, let $g : \Omega \rightarrow \mathbb{C}$ and $f : \Psi \rightarrow \mathbb{C}$ be functions belonging to $\mathcal{H}(\Omega)$ and $\mathcal{H}(\Psi)$, respectively, and such that $f(\Psi) \subseteq \Omega$ and let $z_0 \in \Psi$. Then the composite function $g \circ f$ is in $\mathcal{H}(\Omega)$.

Theorem 1.8 (Derivative of the Inverse Function)

Let Ω be an open subset of $\mathbb{C}$. Suppose $f(z) = w$ is bijective and its inverse function $f^{-1}(w) = z$ is continuous in the open subset $f(\Omega)$, then, if f is differentiable at $z_0 \in \Omega$ and $f'(z_0) \neq 0$, then f^{-1} is differentiable at $w_0 = f(z_0)$ and

$$(f^{-1})'(w_0) = \frac{1}{f'(z_0)}$$

Theorem 1.9 (Differentiation Theorem)

A power series can be differentiated term-by-term within the interval of convergence. In fact, if

$$f(z) = \sum_{n=0}^{\infty} a_n(z - z_0)^n, \quad \text{then} \quad f'(z) = \sum_{n=0}^{\infty} n a_n(z - z_0)^{n-1}$$

for z in the disc of convergence. Both series have the same radius of convergence.

Remark

By repeated application of the preceding result, we conclude that if $k \in \mathbb{N}$, then any analytic function $f(z) = \sum_{n=0}^{\infty} a_n(z - z_0)^n$ can be differentiated term-by-term k times to obtain

$$\sum_{n=k}^{\infty} \frac{n!}{(n-k)!} a_n (z - z_0)^{n-k}.$$

Moreover, this series converges absolutely to $f^{(k)}(z)$ for $|z| < R$ and uniformly over any closed and bounded set in the disc of convergence.

Therefore f is class $C^\infty[D_R(z_0)]$ and for $z = 0$ we obtain the important formula

$$f^{(k)}(0) = k! a_k.$$

Theorem 1.10

If $\sum a_n z^n$ and $\sum b_n z^n$ have the same radius of convergence and converge to the same function f , then

$$a_n = b_n \quad \text{for all } n \in \mathbb{N}.$$

Definition

*Just as every complex number $z = x + iy$ can be characterised by its real and imaginary parts, a complex function f can be specified by two real functions of the complex variable z , namely, $u(z) := u(x, y), v(z) := v(x, y)$ (depending on the real and imaginary parts of z) called the **real** and **imaginary parts** of f , respectively. It is customary to write $w = f(z) = u(x, y) + iv(x, y)$. If $u : \Omega' \rightarrow \mathbb{R}$ and $v : \Omega'' \rightarrow \mathbb{R}$, then $f : \Omega \rightarrow \mathbb{C}$ where $\Omega = \Omega' \cap \Omega''$, that is, the domain of f is the common domain of u and v .*

Definition

Let Ω be an open subset of $\mathbb{C}$, let $z_0 = x_0 + iy_0 \in \Omega$ and let $u : \Omega \rightarrow \mathbb{R}$ be a real-valued function of the complex variable $z = x + iy$. If the function $x \mapsto u(x, y_0)$ is differentiable at x_0 , then the derivative of u at x_0 is called the **first partial derivative of u with respect to x at x_0** , we will denote this number by $\partial_x u(x_0, y_0)$. The **first partial derivative of u with respect to y at y_0** is defined analogously, and will be denoted by $\partial_y u(x_0, y_0)$. In symbols,

$$\partial_x u(x_0, y_0) = \lim_{x \rightarrow x_0} \frac{u(x, y_0) - u(x_0, y_0)}{x - x_0},$$

$$\partial_y u(x_0, y_0) = \lim_{y \rightarrow y_0} \frac{u(x_0, y) - u(x_0, y_0)}{y - y_0}.$$

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Definition

Let $\phi : \Omega \subseteq \mathbb{C} \rightarrow \mathbb{R}$, let $z_0 \in \Omega$ and let $\Phi \subseteq \Omega$. We say that ϕ is **Class** $C^k[\Phi]$ and write $\phi \in C^k[\Phi]$ if the functions $x \mapsto \phi(x, y_0)$ and $y \mapsto \phi(x_0, y)$ are both class $C^k[\Phi]$.

Theorem 1.11 (Cauchy-Riemann Equations)

Let Ω be an open subset of $\mathbb{C}$, let $z_0 = x_0 + iy_0 \in \Omega$ and let $f : \Omega \rightarrow \mathbb{C}$ be differentiable at z_0 . Define maps $u : \Omega \rightarrow \mathbb{R}$ and $v : \Omega \rightarrow \mathbb{R}$ by $u = \Re(f)$, $v = \Im(f)$. Then the partial derivatives of u and v exist at (x_0, y_0) and

$$\partial_x u(x_0, y_0) = \partial_y v(x_0, y_0), \quad \partial_y u(x_0, y_0) = -\partial_x v(x_0, y_0).$$

The latter are known as the **Cauchy-Riemann equations**.

Example 1.12

Show that the function $f(z) = |z|$ is nowhere differentiable. Moreover, show that $g(z) = |z|^2$ is differentiable only at 0 and thus holomorphic nowhere.

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operators**Theorem 1.13**

Let Ω be an open subset of $\mathbb{C}$ and let $u : \Omega \rightarrow \mathbb{R}$ and $v : \Omega \rightarrow \mathbb{R}$ be at least class C^1 at (x_0, y_0) . If these functions satisfy the Cauchy-Riemann equations on Ω , then the function $f : \Omega \rightarrow \mathbb{C}$ defined by $f = u + iv$ is complex differentiable at $z_0 = x_0 + iy_0$.

Remark

We do not have a complete (pointwise) converse for the previous theorem since the Cauchy-Riemann theorem makes no statement regarding the continuity of the partial derivatives.

Definition

Let us define the **Cauchy-Riemann operator**

$$\partial_{\bar{z}} = \frac{1}{2}(\partial_x + i\partial_y),$$

and its conjugate

$$\partial_z = \frac{1}{2}(\partial_x - i\partial_y).$$

Theorem 1.14 (Homogeneous Cauchy Riemann Equations)

Let Ω be an open subset of $\mathbb{C}$, let $f : \Omega \rightarrow \mathbb{C}$ and let $f = u + iv$. If u and v are at least class C^1 , then f is in $\mathcal{H}(\Omega)$ if and only if

$$\partial_{\bar{z}} f(z) = 0 \quad \text{for all } z \in \Omega.$$

In this case we have

$$\partial_z f(z) = f'(z) \quad \text{for all } z \in \Omega.$$

The former is known as the **homogeneous Cauchy-Riemann equation**.

Theorem 1.15

Let Ω be an open and connected subset and $f : \Omega \rightarrow \mathbb{C}$. If f is holomorphic on Ω and $f'(z) = 0$ for $z \in \Omega$, then f is constant on Ω .

Theorem 1.16

Let Ω be an open and connected subset and $f : \Omega \rightarrow \mathbb{C}$. If f is holomorphic on Ω and $|f|$ is constant on Ω , then f is constant on Ω .